

Non-Euclidean Pandrosion IRP

Geodesic Logarithms, Hadamard Charts, and Curvature Terms

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Abstract

The monomial Pandrosion paper shows that the raw map

$$h_{p,y}(s) = 1 - \frac{y-1}{y(1+s+\dots+s^{p-1})}$$

has fixed point $y^{-1/p}$ and exact multiplier

$$\lambda_{p,y} = 1 - \frac{p}{S_p(y^{1/p})}.$$

Its acceleration principle is logarithmic: before applying the raw layer, choose a chart and a scale that make $|\log y|$ small. This note extracts the non-Euclidean content of that principle. On a general Riemannian manifold the expression z^p has no canonical meaning, so a rigorous extension must choose extra structure. The safest extension is the Hadamard/geodesic-dilation model: replace the scalar logarithm by the Riemannian logarithm, replace inversion by geodesic symmetry, and replace scalar scale cells by cells in a tangent logarithmic chart. In that model one renormalized raw Pandrosion layer remains quadratic in the geodesic logarithmic residual, and k layers have dyadic local order 2^k . Hyperbolic space gives an explicit Poincare-ball implementation. Lie groups and symmetric spaces give a richer, noncommutative version in which curvature and Baker–Campbell–Hausdorff commutators enter as higher-order defect terms.

***Main research translation.** Pandrosion IRP does not need Euclidean straight lines; it needs a logarithmic chart in which residuals are measured by a norm, and a chart-scaling rule that keeps this residual norm small before the raw monomial layer acts.*

1 Starting Point From The Monomial Paper

The monomial scheme is organized by three facts:

1. The proportional-mean denominator is

$$S_p(s) = 1 + s + \dots + s^{p-1}.$$

2. The raw multiplier at the normalized target y is

$$\lambda_{p,y} = 1 - \frac{p}{S_p(y^{1/p})}, \quad y \geq 1.$$

It is strictly increasing in $\log y$.

3. Near the ideal target $y = 1$,

$$\lambda_{p,e^\delta} = \frac{p-1}{2p}\delta + O_p(\delta^2).$$

Thus the geometry does not merely normalize magnitudes. It preconditions the multiplier itself. IRP repeats this operation dynamically by recomputing the residual chart after each raw layer.

2 The Necessary Obstruction

Remark 2.1 (No canonical power on a manifold). On a general non-Euclidean manifold M , there is no intrinsic operation $z \mapsto z^p$. A monomial problem therefore needs one of the following extra structures:

- a chosen base point and geodesic dilation, giving $D_p^o(z) = \text{Exp}_o(p \text{Log}_o z)$;
- a group or symmetric-space structure, giving a meaningful group power z^p or a Cartan/geodesic power;
- a concrete model, such as the Poincare ball, where geodesic scalar multiplication has closed form.

Without one of these choices, “Pandrosion for $z^p = x$ ” is not a well-defined mathematical statement.

The rest of this note uses the first option as the rigorous baseline and records the second option as the noncommutative research direction.

3 Hadamard Geodesic-Dilation Model

Let (M, g) be a Hadamard manifold: complete, simply connected, with nonpositive sectional curvature. Then each exponential map $\text{Exp}_o : T_o M \rightarrow M$ is globally well behaved in the Cartan–Hadamard setting. The positive multiplicative line used by the original paper is the flat special case after the coordinate change $y = e^\xi$.

Definition 3.1 (Geodesic monomial). Fix $o \in M$ and $p \geq 2$. Define the geodesic p -dilation

$$D_p^o(z) = \text{Exp}_o(p \text{Log}_o z).$$

The Hadamard monomial root problem is

$$D_p^o(z) = x.$$

If $X = \text{Log}_o x$, its exact root is

$$r = \text{Exp}_o(X/p).$$

This problem is deliberately narrow, but it is the cleanest non-Euclidean analogue of the logarithmic root problem. It replaces $\log y$ by the vector $X = \text{Log}_o x$.

Definition 3.2 (Raw scalar Pandrosion length layer). For a residual length $\delta \geq 0$, define

$$s_p(\delta) = h_{p, e^\delta}(1) = 1 - \frac{1 - e^{-\delta}}{p},$$

and

$$\Theta_p(\delta) = \log(e^\delta s_p(\delta)^{p-1}) = \delta + (p-1) \log s_p(\delta).$$

The number $\Theta_p(\delta)$ is the logarithmic correction produced by one raw Pandrosion layer from the neutral point in the residual chart.

Lemma 3.3 (Scalar expansion). As $\delta \rightarrow 0^+$,

$$\Theta_p(\delta) = \frac{\delta}{p} + \frac{(p-1)^2}{2p^2} \delta^2 + O_p(\delta^3).$$

Consequently,

$$\delta - p\Theta_p(\delta) = -\frac{(p-1)^2}{2p} \delta^2 + O_p(\delta^3).$$

Proof. Use

$$s_p(\delta) = 1 - \frac{\delta}{p} + \frac{\delta^2}{2p} + O_p(\delta^3)$$

and expand $\log(1+t)$. This gives

$$\log s_p(\delta) = -\frac{\delta}{p} + \frac{p-1}{2p^2} \delta^2 + O_p(\delta^3),$$

which proves both displayed estimates. □

Definition 3.4 (Continuous Hadamard IRP layer). Let $u = \text{Exp}_o(U)$ be the current root-coordinate approximation and put

$$A(u) = X - pU \in T_o M, \quad \delta(u) = \|A(u)\|_g.$$

If $\delta(u) > 0$, one continuous-scale Hadamard IRP layer is

$$U_+ = U + \Theta_p(\delta(u)) \frac{A(u)}{\delta(u)}, \quad \mathcal{I}_p^o(u) = \text{Exp}_o(U_+).$$

If $\delta(u) = 0$, set $\mathcal{I}_p^o(u) = u$.

Theorem 3.5 (Quadratic residual collapse in the Hadamard log chart). For the geodesic-dilation problem $D_p^o(z) = x$, the continuous Hadamard IRP layer satisfies

$$\|A(\mathcal{I}_p^o(u))\|_g = \frac{(p-1)^2}{2p} \|A(u)\|_g^2 + O_p(\|A(u)\|_g^3).$$

Equivalently, if the root logarithmic error is

$$E(u) = U - X/p,$$

then

$$\|E(\mathcal{I}_p^o(u))\|_g = \frac{(p-1)^2}{2} \|E(u)\|_g^2 + O_p(\|E(u)\|_g^3).$$

Proof. Since $A(u) = X - pU = -pE(u)$, the update changes A by

$$A_+ = X - pU_+ = A - p\Theta_p(\|A\|) \frac{A}{\|A\|}.$$

Taking norms and applying the scalar expansion gives the result. $\square$

Corollary 3.6 (Dyadic local order). *For every fixed $k \geq 1$, the k -fold local composition satisfies*

$$\|E((\mathcal{I}_p^o)^{\circ k}(u))\|_g = O_{p,k}(\|E(u)\|_g^{2^k}).$$

Proof. The previous theorem gives $\|E_+\| \leq C\|E\|^2$ in a sufficiently small neighborhood of the root. Iteration gives the displayed dyadic order. $\square$

4 Geodesic Chart Scaling

The continuous layer above assumes the residual length is already small. The non-Euclidean analogue of Thales/Riemann scaling is to reduce the residual inside T_oM before applying the raw layer.

Definition 4.1 (Logarithmic scale palette in a tangent chart). Let $\mathcal{L} \subset T_oM$ be a finite or countable scale palette. For a residual $A = X - pU$, choose $B \in \mathcal{L}$ and work with the normalized residual

$$A_B = A - pB.$$

The chart-scaled layer is

$$U_+ = U + B + \Theta_p(\|A_B\|) \frac{A_B}{\|A_B\|},$$

with the evident convention when $A_B = 0$.

The geodesic inversion analogue is the symmetry at the chart center:

$$\sigma_o(\text{Exp}_o V) = \text{Exp}_o(-V).$$

Thus the direct chart sees A , and the inverted chart sees $-A$. Selecting the chart and scale means minimizing $\|A_B\|$ or $\| -A - pB \|$ among admissible choices.

Proposition 4.2 (Radial two-chart lattice bound). *Fix a geodesic ray and a root-scale spacing $h = \log q > 0$. Let admissible scales on that ray be $B_m = mh\nu$, $m \in \mathbb{Z}$, where ν is a unit tangent direction. For a radial residual $A = a\nu$, the best direct scale leaves a remainder $r \in [0, ph)$, while allowing the geodesic symmetry leaves a remainder*

$$r_* = \min(r, ph - r).$$

Hence the worst normalized residual length is at most $ph/2$, and the exact raw multiplier is bounded by

$$1 - \frac{p}{S_p(e^{h/2})}.$$

With K equally spaced offsets on the same logarithmic period, the sharp worst-case bound becomes

$$1 - \frac{p}{S_p(e^{h/(2K)})} = 1 - \frac{p}{S_p(q^{1/(2K)})}.$$

Proof. This is the same circle-covering proof as in the monomial paper, now applied to the scalar coordinate a along the geodesic ray. The multiplier bound follows from the monotonicity of λ_{p,e^δ} in δ and the identity $\lambda_{p,e^\delta} = 1 - p/S_p(e^{\delta/p})$. $\square$

5 Hyperbolic Poincare-Ball Realization

The most concrete non-Euclidean model is the Poincare ball

$$\mathbb{B}^n = \{x \in \mathbb{R}^n : \|x\| < 1\}$$

with curvature -1 and center 0 . Its radial exponential and logarithm are

$$\text{Exp}_0(V) = \tanh(\|V\|/2) \frac{V}{\|V\|}, \quad \text{Log}_0(x) = 2 \operatorname{artanh}(\|x\|) \frac{x}{\|x\|}.$$

Therefore geodesic scalar dilation is

$$D_a^0(x) = \text{Exp}_0(a \text{Log}_0 x) = \tanh(a \operatorname{artanh} \|x\|) \frac{x}{\|x\|}.$$

This is the Mobius/gyrovector scalar multiplication used in hyperbolic models.

Example 5.1 (Hyperbolic Pandrosion IRP layer). Given target $x \in \mathbb{B}^n$ and current $u \in \mathbb{B}^n$:

1. Compute $X = \text{Log}_0 x$ and $U = \text{Log}_0 u$.
2. Set $A = X - pU$ and $\delta = \|A\|$.
3. Optionally choose a tangent scale B and replace A by $A - pB$.
4. Update

$$U_+ = U + B + \Theta_p(\|A - pB\|) \frac{A - pB}{\|A - pB\|}.$$

5. Return $u_+ = \text{Exp}_0(U_+)$.

This solves the hyperbolic geodesic root problem $D_p^0(u) = x$, not the Euclidean coordinate equation $u^p = x$.

6 Beyond The Radial Model: Curvature And Noncommutativity

The Hadamard construction above is rigorous because it fixes a logarithmic chart. It becomes genuinely curved when the chart center moves, or when the monomial power is a group/symmetric-space operation rather than a radial geodesic dilation.

Definition 6.1 (Lie-group residual IRP, local form). Let G be a Lie group with exponential map $\exp : \mathfrak{g} \rightarrow G$. For the local group-power problem $z^p = x$, define a right residual

$$R(u) = u^{-p}x, \quad A(u) = \log R(u) \in \mathfrak{g}.$$

A residual Pandrosion layer applies the scalar length correction in the residual direction:

$$u_+ = u \exp\left(\Theta_p(\|A(u)\|) \frac{A(u)}{\|A(u)\|}\right),$$

with scale and inversion choices made in $\mathfrak{g}$.

Remark 6.2 (Where curvature appears). In the abelian case, the preceding formula reduces to the scalar logarithmic IRP. In the noncommutative case,

$$\log(\exp A \exp B) = A + B + \frac{1}{2}[A, B] + \dots$$

introduces Baker–Campbell–Hausdorff commutators. On symmetric spaces the same role is played by curvature/Jacobi-field terms. Thus the local research target is not the scalar quadratic term, which is already known, but the estimate

$$\begin{aligned} \|A_+\| &\leq C_p \|A\|^2 + C_{\text{curv}} \|A\| \|[A, U]\| \\ &\quad + O(\|A\|^3). \end{aligned}$$

If the residual direction nearly commutes with the current chart, or if the problem is radial in a symmetric space, the commutator term vanishes and the Hadamard theorem above is recovered.

Proposition 6.3 (Local curvature-stable template). *Let a moving-chart non-Euclidean IRP layer be written in normal coordinates at a root r as*

$$E_+ = Q_p(E) + C_\kappa(E),$$

where $Q_p(E)$ is the flat Pandrosion residual map with $\|Q_p(E)\| \leq c_p \|E\|^2$ and the curvature defect satisfies $\|C_\kappa(E)\| \leq c_\kappa \|E\|^3$ on a normal ball. Then the layer remains quadratic:

$$\|E_+\| \leq C \|E\|^2.$$

Consequently fixed k -fold IRP compositions still have local order 2^k .

Proof. For $\|E\|$ sufficiently small, the cubic defect is dominated by the quadratic term. The dyadic-order argument is the same as in the flat and Hadamard settings. $\square$

7 Research Program

1. **Formal baseline.** Prove the Hadamard geodesic-dilation theorem in Lean or another proof assistant. This requires only normal coordinates, vector norms, and the scalar expansion of Θ_p .
2. **Hyperbolic implementation.** Implement the Poincare-ball layer with Exp_0 , Log_0 , and tangent scale palettes. Verify exact reduction to the monomial paper on every geodesic ray.

3. **Spherical local model.** Repeat the construction on the sphere only inside injectivity-radius balls. The cut locus replaces the global Hadamard guarantee, so atlas switching is mandatory.
4. **Symmetric positive-definite model.** Use the affine-invariant SPD geometry, where logarithms and geodesics are explicit. Radial spectral problems are exact; noncommuting updates measure the first serious curvature defect.
5. **Lie-group BCH bounds.** Quantify when the noncommutative group residual keeps the same quadratic IRP order. The key estimate is the size of the commutator defect relative to the scalar Pandrosion residual collapse.

8 Conclusion

The non-Euclidean version of Pandrosion IRP should not be phrased as a naive replacement of straight lines by curved lines. The correct invariant object is the logarithmic residual. In the original monomial paper this residual is the number $\log y$ on the positive multiplicative line. In a Hadamard manifold it becomes $\text{Log}_o x - p \text{Log}_o u$. Chart scaling means minimizing the norm of that residual before the raw layer acts. Under this translation, the local IRP mechanism survives unchanged: one renormalized raw layer is quadratic, and iterated layers have dyadic order. Curvature enters only when the chart moves or when the underlying power law is noncommutative; those terms form the next research problem.

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